Weekly Site Report

BLACK ROCK WEEKLY REPORT

Weekly PSD Report

EPIROC SA

Contents

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Weekly Service Report – Black Rock Mine Operations

Report type:	**Weekly**	**Submitted to:**	**Sello Taku and Sipho**
Week Date:	13 February 2026	**End Date**	19 February 2026
Reported By:	Philip Moller		
BEV:			
Service Tech:	Ivan Phiri	**Service Tech:**	Mathews Mosemane
Service Tech:	Patricia Bulelwa	**Service Tech Assistant:**	Patrick Yoko (Batteries)
Specialist:	Frikkie van der Walt (Batteries)		
Field Service:			
Service Tech:	Lucas Slaffa	**Service Tech:**	Phineas Sibanda
CAS Project:			
Team Leader:	Wisani Shilenge	**Service Tech:**	Phineas Sibanda
Service Tech:	Julius Manamela	**Service Tech:**	Joseph Mampa
Certiq Project:			
Team Leader:	Segunda Fernado	**Service Tech:**	
Mobilaris:			
Team Leader:	Rupert Grobler		

Report Summary

General Summary

The purpose of this report is to give an overview of all the Epiroc activities at Blackrock Mine Operations (BRMO).

BEV - Critical Breakdown Summary

- DT171 - A-frame that was previously damaged failed again.
- Battery breakdowns increased. Currently busy with TMS hyper care. Most of the breakdowns are TMS related. We are having coolant pump failures. New pumps ordered and will arrive 07 March. Changing coolant and coolant hoses on services.

- DT172 – Need to commission machine.
- FL112 – Re-occurring Strata faults
- I looked at the job cards on click view to see if I can get more info and do root cause analysis on the machines and look for re-occurring faults or trends. Unfortunately, it seems like job cards are not being filled in properly anymore. We are left with limited reporting to do this kind of RCAs.

CAS L9 Implementation - Critical Issues Summary

- Lab scale tests with new software currently underway.
- Invoicing and sign off to be completed.
- MH4.0 software to be loaded on the BEV machines. Currently busy with the lab scale tests and approval of the software.

Certiq & Mobilaris - Critical Issues Summary

Passport 360 Compliance

- BEV Technicians: **96%**
- CASL9 (Gloria): **99%**
- CASL9 (Nch2): **95%**
- CAS L9 (Nch3): **98%**
- Certiq & Mobilaris N2: **99%**
- Certiq & Mobilaris Gloria: **88%**

BEV Report

Weekly machine utilization summary:

Friday 13 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 - Night shift: Auto electrical breakdown – Oil level sensor, No feedback. (12H18min)
4. FL 108 -
5. FL 112 – Night shift - 3rd Party Supplier Fault - STRATA (1H59min)
   Morning shift - 3rd Party Supplier Fault – STRATA (4H17min)
   Afternoon shift - 3rd Party Supplier Fault – STRATA (6H31min)
6. FL 113 - Morning shift: Mechanical breakdown – No feedback. (1H04min)
7. DT 146 - Morning shift: Mechanical breakdown – No feedback (3H)
8. DT 147 -
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 - Ongoing breakdown – A frame needs to be replaced
14. DT 172 -

Saturday 14 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 - Morning shift: Boilermaker breakdown – Half arrows. (2D13H42min)
5. FL 112 - Morning shift - 3rd Party Supplier Fault – STRATA (1H59min)
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 - Morning shift: Mechanical breakdown – No feedback. (1H11min)
13. DT 171 - Ongoing breakdown – A frame needs to be replaced
14. DT 172 -

Sunday 15 February 2026:

1. FL 98 -
2. FL 99 - Morning shift: Electrical breakdown – Fixed a hydraulic oil leak. (17min)
3. FL 107 -
4. FL 108 -
5. FL 112 - Morning shift - 3rd Party Supplier Fault – STRATA (55min)
6. FL 113 -
7. DT 146 - Morning shift: Mechanical breakdown – No feedback. (3H33min)
8. DT 147 -
9. DT 149 –
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 - Ongoing breakdown – A frame needs to be replaced
14. DT 172 -

Monday 16 February 2026:

1. FL 98 - Morning shift: Mechanical breakdown – Changed transmission rubbers. (1D23H27min)
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 - Morning shift - 3rd Party Supplier Fault – STRATA (1H59min)
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 –
10. DT 150 -
11. DT 162 -

12. DT 163 - Morning shift: Auto electrical breakdown – Was standing with HVIL fault (VPX00015) at squares, recovered machine to the workshop and changed battery. (3H33min)
13. DT 171 - Ongoing breakdown – A frame needs to be replaced
14. DT 172 – Afternoon shift: Auto electrical breakdown – No feedback. 12H31min)

Tuesday 17 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 - Night shift: Electrical breakdown – No feedback. (45min)
5. FL 112 – Afternoon shift: Mechanical breakdown – No feedback. (1H07min)
6. FL 113 -
7. DT 146 -
8. DT 147 - Afternoon shift: Auto electrical breakdown – No feedback. (13min)
9. DT 149 -
10. DT 150 - Afternoon shift: Electrical breakdown – BMS error, Isolation fault on battery VPX00036. (2H14min)
    Night shift: Mechanical breakdown – No feedback. (1H)
11. DT 162 -
12. DT 163 -
13. DT 171 - Ongoing breakdown – A frame needs to be replaced
14. DT 172 -

Wednesday 18 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 - Night shift: Electrical breakdown – No feedback. (32min)
8. DT 147 - Night shift: Electrical breakdown – Replaced e-stop relays, Took relays from DT172. (1D11H13min)
9. DT 149 – Morning shift: Auto electrical breakdown –Traction motor shows high temp, reset machine. (57min)
    Night shift: Electrical breakdown – No feedback. (5min)
10. DT 150 -
11. DT 162 - Morning shift: Auto electrical breakdown – No feedback (34min)
    Night shift: Electrical breakdown – No feedback. (8H46min)
12. DT 163 - Night shift: Electrical breakdown – No feedback. (56min)
    Night shift: Auto electrical breakdown – Battery not connecting, recovered machine. (10H05min)
13. DT 171 - Ongoing breakdown – A frame needs to be replaced
14. DT 172 -

Thursday 19 February 2026:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 – Afternoon shift: Mechanical breakdown – Centre bearing bolts loose, tightened bolts. (2H57min)
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 - Morning shift - 3rd Party Supplier Fault – STRATA. (1H07min)
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 - Ongoing breakdown – A frame needs to be replaced
14. DT 172 – Morning shift: Auto electrical fault – HVIL fault. Waiting new HVIL cables

Planned work for the coming week:

NCH3:

- Inspect HD0062 to see what need to be replaced on articulation. Will compile what parts will be needed and order them before machine go out for repairs.
-

NCH2:

- Fault finding on FL0122, machine CAN fault after dummy cab was installed, busy with fault finding. Neet to commission machine.
- Commission RT0061 (New close cab machine)

Gloria

- Safety systems audit to be done. 2 Machines left.
- Audit of RT0045 and RT0046. Machines have low availability.

Improvement and actions:

1. We need to look at the data of the charger interrupts. This function was tasked with Desmond. We will start seeing data coming through this week.

2. Post 3,6 and 7 cables replaced, Piet to put in an order for all other chargers.

3. Auxiliary motor spline grease TSNB not done. Only DT147 completed. We need to schedule the machines to complete this campaign. Grease already on site.. Piet and myself to schedule machines. We started the discussion.
4. Regeneration knob set to 100% completed. Need to sign off change management and attendance register for operators and artisans underground.

5. Need to physically check status of chargers and compare to report. Chargers checked. All modules put back except for charger 2 with a faulty module.
6. VCA, VCB cables and discharge resistors identified machines to be replaced. Parts partially ordered. Followed up from buyers, some orders still in approval phase. - Wimpie and Piet, can you please update me on what machines resistors and connectors was replaced.

DT147

- Discharge resistor not working
- VCA connector worn
- VCB HVIL pins pushed back
- VCB pins pushed back
- VCB connector clamps broken

DT149

- Discharge resistor not working
- VCA connector worn

DT 150

- Main switch need to be replaced
- VCB insulator missing
- VCB pins have Arc marks
- VCA connector worn
- VCA cable trunking loose

DT171

- Discharge resistor not working
- VCB connector insulator missing

FL107

- Discharge resistor not working
- VCA plug worn
- VCB insulator missing
- VCB connector clamps broken

FL98

- Discharge resistor not working
- VCA connector worn
- VCB insulators missing

FL108

- VCA connectors joint inside plug, need to be replaced
- VCA plug worn
- Insulators missing

7. Audits to be done on all A frame retaining bolts of dump trucks. Could not do this. Currently busy with Brake wear measurements on BEV machines.

DT147

Right rear – 3.28mm
Left rear – 3.18mm
Front Right – 3.4mm
Front left – 3.12mm

DT172
Right rear – 4.72mm
Left rear – 4.72mm
Front Right – 4.76mm
Front left – 4.7mm

DT150
Right rear – 2.9mm
Left rear – 2.8mm
Front Right – 2.5mm
Front left – 2.1mm

TSNB completion percentile

B4, B5 and chargers

B4 Campaign	
Campaign	**Percentile**
DCDC	95%
MH 6.3	80%
TMS SW V2.0.3.0	20%

Charger SW Campaign	
SW V3.4.7	90%

B5 Campaign	
Campaign	**Percentile**
DCDC	93.33 %
MH 6.3	0%
TMS V2.0.3.0	0%

8.
9. Busy with TMS hyper care on all batteries.

Services

Nchwaning 3

Maintenance Schedule Feb 2026				
Friday	Monday	Tuesday	Wednesday	Thursday
20-Feb-26	23-Feb-26	24-Feb-26	25-Feb-26	26-Feb-26
DT0118	DT0119	DT0131	DT0161	
	FL0108		FL0098	
HD0049	HD0050	HD0051	HD0052	HD0055
RT0057	RT0058	RT0025	RT0044	RT0047
SR0038	SR0039	SR0043	SR0044	SR0024
	UV0126	UV0131	UV0134	UV0135
	UV0107	UV0112	UV0122	UV0132
UV0079	UV0081	UV0082	UV0086	UV0088
		UV0091	UV0094	
VPY-00083 (BY0026)	VPX-00028(BY0009)	VPX-00028(BY0009)	VPX-00031(BY0018)	VPY-00088(BY0022)
Highlighted in Blue Involve STRATA Sats				

Battery and Charger Status Report

General summary

ST14 – B4 battery packs

- We currently have 10 x ST14 – B4 battery packs and 6 x ST14 machines operational underground. This is equal to the committed ratio of 1.6 batteries per machine.

MT42 – B5 battery packs

- We currently have 12 x MT42 – B5 battery packs operational underground. This is above the committed ratio of 1.6 batteries per machine.

160 kW Chargers

General

- DCDC campaign 95% complete. Initial indication from team is that charger stops significantly reduced.
- TSNB software V6.3 to be installed on batteries. (level 3 safe state)
- Fleet number stickers ordered for batteries for better identification.

Table 1: Battery pack availability for week 2025

Battery Type	Battery ID	Status	Comment
B4 - ST14	VPY-00011	**Working**	16 Feb: Coolant pump failed – Swap TMS with VPY00076. (1D)
B4 - ST14	VPY-00031	**Working**	17 Feb: Repaired HV cable between TMS and battery. (1H)
B4 - ST14	VPY-00051	**Breakdown**	19 Feb: Replaced leaking coolant hose and re-pressurized system. (2H)
B4 - ST14	VPY-00048	**Working**	16 Feb: VPC#4 safe state, replaced sub pack. (3D) 17 Feb: CAN fault – repaired diagnostic plug pins. (1D04H)
B4 - ST14	VPY-00049	**Working**	
B4 - ST14	VPY-00088	**Working**	
B4 - ST14	VPY-00086	**Working**	
B4 - ST14	VPY-00076	**Breakdown**	
B4 - ST14	VPY-00083	**Working**	
B4 - ST14	VPY-00041	**Working**	
B5 - MT42	VPX-00016	**Breakdown**	VPC#3 is in a safe state. Waiting for Scania. (9days)
B5 - MT42	VPX-00015	**Working**	
B5 - MT42	VPX-00023	**Working**	
B5 - MT42	VPX-00017	**Working**	
B5 - MT42	VPX-00010	**Working**	
B5 - MT42	VPX-00050	**Working**	16 Feb: VPC isolation fault – Replaced sub pack. (3D)
B5 - MT42	VPX-00031	**Working**	
B5 - MT42	VPX-00036	**Working**	16 Feb: Replaced coolant hose and pressurized system. (3H) 17 Feb: VPC #3 low isolation fault – Replaced subpack. (2D)
B5 - MT42	VPX-00028	**Working**	
B5 - MT42	VPX-00026	**Working**	
B5 - MT42	VPX-00024	**Working**	
B5 - MT42	VPX-00048	**Working**	
B5 - MT42	VPX-00044	**Underground**	

Battery charger summary:

Charger 1

Module 2 switch off and pulled out. All fans working on the 3 modules that is in used. Switched on.

Charger 2
Module 2 switch off and pulled out. All fans working on the 3 modules that is in used. Module 2 faulty.

Charger 3
Module 4 switch off and pulled out. All fans working on the 3 modules that is in used. Switched on.

Charger 7
Module 4 switch off but still in place. All fans working on the 3 modules that is in used. Switched on.

Charger 4
Module 1switch off and pulled out. All fans working on the 3 modules that is in used. Switched on.

Charger 5
Module 4 switch off and pulled out. All fans working on the 3 modules that is in used. Switched on.

Charger 6
Module 1 switch off and pulled out. All fans working on the 3 modules that is in used. Switched on.

Charger 8
Module 4 switch off and pulled out. All fans working on the 3 modules that is in used. Switched on.

FST Report Summary

CAS Report Summary

Report Type:	Weekly	**Project Leader:**	Hennie van Niekerk
Start Date:		**Report By:**	
End Date:		**Contract No:**	BPE145D

Critical Issues on Batteries

- VPX00048 – Shows overheat. B6 sensor plug had a loose connection
- Vpy00011 – Overheat. Water pump failure. Changed TMS.

- VPY00051 – Temps running high. Caught the issue before battery overheat. Changed 2 coolant hoses and busy looking for AC gas leak.
- VPX00016 - (Last week) Battery have all the sub packs in that Scania must unlock. Will continue with fault finding once that is done.

Progress Update